Jahangirnagar University

Department of Computer Science and Engineering

2^{nd} Year 1^{st} Semester B.Sc. (Hons.) Final Examination -2021

Course Title: Electronic Device and circuit Theory-II
Time: 3 hours

Course No: **CSE 207**
Full Marks: **60**

[Answer each of the following questions. Figures in the right margin indicate marks.]

1.

Question no 1 is based on CO1. Answer *All of them.*

a) Mention the works of zero drift amplifier. 2
b) State the meaning of slew rate. 2
c) Why do we need virtual ground? 2
d) Identify the causes of amplifier distortion. 2
e) How can we find the Dark current? 2
f) Identify Common-Mode Operation and Common-Mode Rejection of an OpAmp. 2

2.

Question no 2 is based on CO2. Answer *Any Three* out of *Four*.

a) Formalize the principles of Transformer-Coupled Push-Pull Class B Amplifier. 4
b) Write in your own words how the construction of the hot carrier diode is significantly different from the conventional semiconductor diode. In addition describe its mode of operation. 4
c) Draw and describe the operation and applications of LCD. 4
d) Design a astable multivibrator and describe its configuration. 4

3.

Question no 3 is based on CO3. Answer *Any Three* out of *Four*.

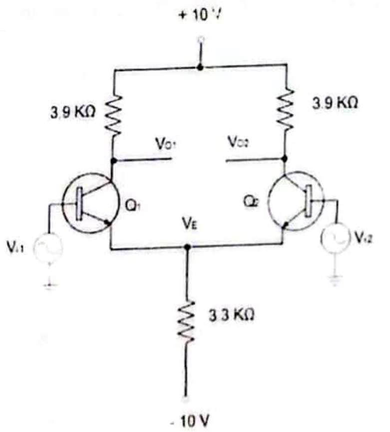


Figure: 3(a)

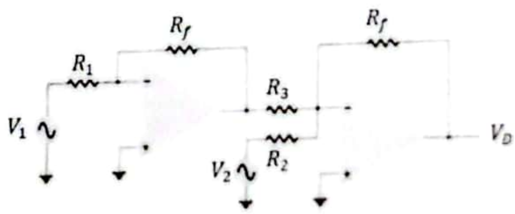


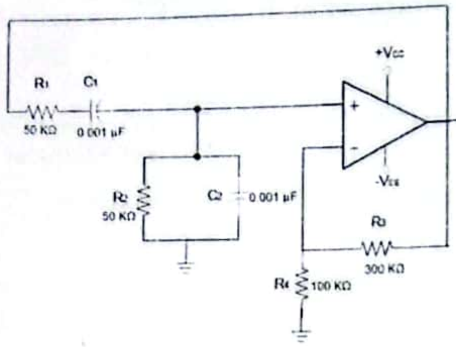
Figure: 3(b)

a) Calculate the DC voltage and currents for the given circuit in **figure 3(a)**. 4
b) Illustrate the output voltage of the given circuit in **figure 3(b)**. 4
Determine the output for the circuit with components $R_f = 1\ MOhms, R_1 = 1000\ KOhms, R_2 = 50\ KOhms\ and\ R_3 = 500\ KOhms$ consider $V_2 = 2V\ and\ V_1 = 1\ V$

c) Calculate the efficiency of a transformer-coupled class A amplifier for a supply of 12 V and outputs of: $i)\ V(p) = 12V, ii)\ V(p) = 6V\ and\ iii)\ V(p) = 2$ 4

d) For an op-amp having a slew rate of SR = 5 V/ms, what is the maximum closed-loop voltage gain that can be used when the input signal varies by 0.5 V in 10 ms? 4

Question no 4 is based on CO4. Answer *Any Three* out of *Four*.



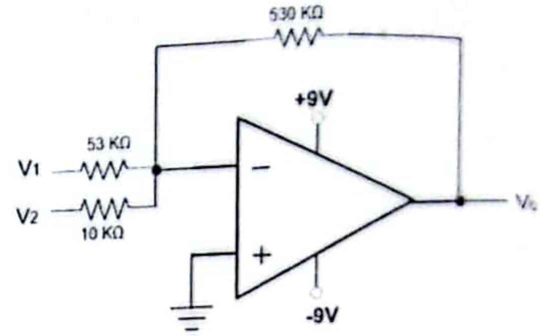
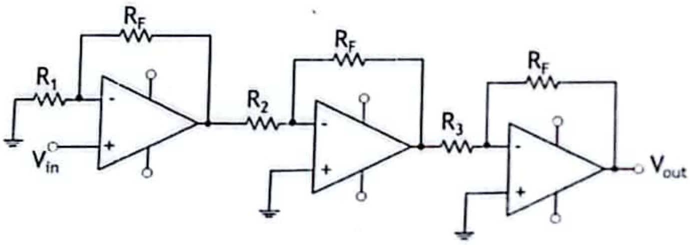


Figure: 4(a) **Figure: 4(b)**

4\. a) Calculate the resonant frequency of the Wien bridge oscillator in Figure 4(a). 4

b) Calculate the output voltage for the circuit when the inputs are $V_1 = 50\ mV\ sin\ (1000\ t\ )$ and $V_2 = 10\ mV\ sin(3000\ t\ )$ in Figure 4(b). 4

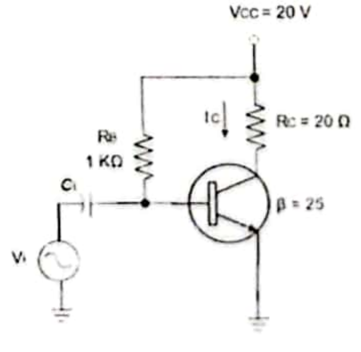
c) Calculate the output voltage using the circuit for resistor components of value $R_F = 47\ k\Omega, R_1 = 4.3\ k\Omega, R_2 = 33\ k\Omega, and\ R_3 = 33\ k\Omega$ for an input of $40\ mV$. 4



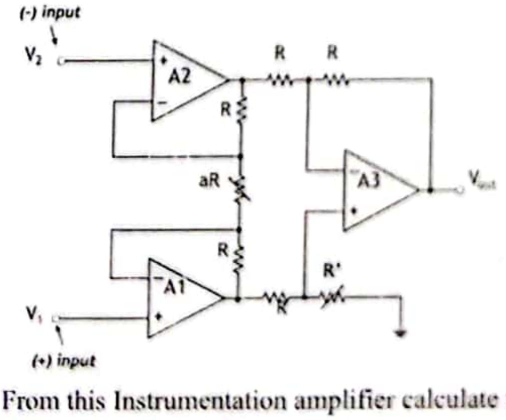
d) How can we design a Quasi-Complementary Push-Pull Amplifier with proper indication? 4

Question no 5 is based on CO5. Answer *Any Two* out of *Three*.

5\. a) Calculate the Input power, Output power, and Efficiency of the amplifier circuit where an input voltage that results in a base current of $20\ mA$ peak. 6



b) 6



From this Instrumentation amplifier calculate the voltage gain, if $R = 20\ k\Omega$ and $aR = 40\ \Omega$.

c) For a VCO, prove that free-running or center operating frequency f_o can be expressed by 6

$$f_0 = \frac{2}{R_1 C_1}\left(\frac{V^+ - V_C}{V^+}\right)$$

Jahangirnagar University
Department of Computer Science and Engineering
2^{nd} Year 1^{st} Semester B.Sc. (Hons.) Final Examination -2021

Course Title: Mathematics III (Vector, Complex Variable, Fourier Analysis and Laplace Transformation) **Time:** 3 Hours
Course No: MATH 201 **Full Marks:** 60

Each section carries equal marks. Figures in the right margin indicate marks.
Answer any Six (06) questions out of Eight (08) questions.

1. a. Define the gradient of a Scalar, the Divergence of a vector, and the curl of a vector. If $\phi(x,y,z) = 3x^2y - y^3z^2$ find $\underline{\nabla}\phi$ at the point $(1,-2,-1)$. 5

   b. Show that $\underline{\nabla} r^n = nr^{n-2}\underline{r}$, where $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$. 5

2. a. State Green's theorem, Stokes theorem and the Gauss divergence theorem. 5

   b. Find the total work done in moving a particle in a force field given by $\underline{F} = 3xy\underline{i} - y^2\underline{j}$ along the curve $y = 2x^2$. 5

3. a. If $f(z)$ is analytic at every point on and within the simply closed curve C where a is within C then show that 4

$$\int_C \frac{f(z)}{z-a}dz = 2\pi i\, f(a)$$

   b. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z^2 - 9z + 20}dz$; where $C: z = 6e^{i\theta}; 0 \le \theta \le 2\pi$. 3

   c. Evaluate $\oint_C \frac{e^{2z}}{(z-1)^4}dz$; where C is the circle $|z| = 3$. 3

4. a. State and proof Taylors Theorem. 5

   b. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series valid for: 5

      i. $|z| > 3$
      ii. $0 < |z+1| < 2$
      iii. $|z| < 1$

5. a. Find the residues of $f(z) = e^z \operatorname{cosec}^2 z$ at all its poles in the finite plane. 3

   b) Evaluate $\int_0^\infty \frac{dx}{x^4 + a^4}$; $a > 0$. 5

   c) State Laurent Series. 2

6. a. Find the Fourier series expansion of $f(x)=\begin{cases} x-\pi, & -\pi<x<0 \\ \pi-x, & 0<x<\pi \end{cases}$. 5

b. Find the Fourier Transform of $f(x)=\begin{cases} 1-x^2, & |x|<1 \\ 0, & |x|>1 \end{cases}$. 5

7. a. How do you define an analytic function? Obtain Cauchy-Riemann equations. Express, giving all detail, these equations in polar form. 5

Determine whether $u(x,y)=e^{-2xy}\,Sin(x^2-y^2)$ is harmonic or not. If so, find the conjugate harmonic function of u. 5

8. a. Define Laplace transform. Find the Laplace transform of the function $F(t)=t^n e^{at}$. 4

b. Solve the following differential equations by using Laplace transform 6

i. $\dfrac{dx}{dt}+5x=e^{2t}, \qquad x(0)=1$

ii. $\dfrac{d^2y}{dt^2}+y=3\sin(2t), \qquad y(0)=3, y'(0)=1.$

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Good Luck!!!

Jahangirnagar University

Department of Computer Science and Engineering

2^{nd} Year 1^{st} Semester B.Sc. (Honors) Final Examination - 2021

Course Title: Numerical Methods
Time: 3 Hours

Course No: **CSE-205**
Full Marks: **60**

[Answer each of the following questions.
Each question carries equal marks. Figures in the right margin indicate marks.]

1.

Section-I (CO1)

(Answer all the questions.)

a) Define the round-off and truncation errors with appropriate examples. 2

b) Define interpolation. List out the methods available for interpolation. 2

c) Differentiate between Jacobi iteration method and Gauss-Seidel method. 2

d) Define regression and explain its necessary. 2

e) Explain the necessity of using numerical techniques to obtain the estimates of function derivatives. 2

f) Write down the order of the error for the Trapezoidal and Simpson's method of numerical integration. 2

2.

Section-II (CO2)

(Answer Any Three out of Four)

a) Illustrate with the help of a block diagram the process of numerical computing. 4

b) Explain that $f(x) = x^4 + x - 1$ has a real root α in the interval [0.5, 1.0]. 4

c) Demonstrate Matrix Inversion Method for solving the following system of equations using: 4

$$\begin{aligned} 2x_1 + 4x_2 - 6x_3 &= -8 \\ x_1 + 3x_2 + x_3 &= 10 \\ 2x_1 - 4x_2 - 2x_3 &= -12 \end{aligned}$$

d) Demonstrate Taylor method for solving the following differential equation: 4

$$y' = x^2 + y^2$$

where, x = 0.25 and x = 0.5, given y(0) = 1.

3.

Section-III (CO3)

(Answer Any Three out of Four)

a) Show that Newton's method for the scalar equation f(x) = 0 is quadratically convergent (assuming that f(x) is sufficiently smooth, and that f'(x) ≠0 at the root). 4

b) Calculate the area under the curve using two Segment Trapezoidal Rule. 4

$$f(x) = \frac{300x}{1+e^x} \quad from \;\; x = 0 \;\; to \;\; x = 10$$

c) Use the Secant method to estimate the root of the equation: $f(x)=x^2-4x-10$ with the initial estimates of $x_1=4$ and $x_2=2$. Show at least four iterations. 4

d) The values of distance travelled by a car at various time intervals are given in the following table. Estimate the velocity at time $t = 5$, $t = 7$ and $t = 9$. 4

Time, t(s)	5	6	7	8	9
Distance travelled, s(t) (km)	10.0	14.5	19.5	25.5	32.0

4.

Section-IV (CO4)

(Answer Any Three out of Four)

a) Solve the following system of equations by Gauss-Seidel iteration method (show at least four iterations): 4

$$\begin{aligned} x_1 + 2x_2 - 3x_3 &= 4 \\ 2x_1 + 4x_2 - 6x_3 &= 8 \\ x_1 - 2x_2 + 5x_3 &= 4 \end{aligned}$$

b) Deduce the following Trapezoidal rule for approximating $\int_a^b f(x)dx$. 4

$\int_a^b f(x)dx = \frac{h}{2}[f(a) + f(b)] - \frac{h^3}{12} f''(\xi)$, where h = b − a.2

c) Classify the four possible solution conditions of a system of linear equations. Analyze each one of them with proper illustration. 4

d) Examine shooting method to solve the following equation: 4

$$\frac{d^2y}{dx^2} = 6x, \qquad y(1) = 2 \quad and \quad y(2) = 9$$

in the interval (1, 2).

5.

Section-V (CO5)

(Answer Any Two out of Three)

a) The velocity of a rocket is given by 6

$$v(t) = 2000\ln\left[\frac{14\times10^4}{14\times10^4 - 2100t}\right] - 9.8t, \quad 0 \le t \le 30$$

Where v is given in m/s and t is given in seconds.

i. Use forward difference approximation of the first derivative of v(t) to calculate the acceleration at t = 16s. Use a step size of $\Delta t = 2$s

ii. Find the exact value of the acceleration of the rocket.

b) Evaluate the least square regression to fit a straight line to the following set of data. 6

x:	1	3	4	6	8	9	11
y:	1	2	4	4	5	7	8

c) Consider the tri-diagonal matrix, 6

$$\begin{pmatrix} 4 & 2 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

To find eigenvalues one uses a QR algorithm involving successive iterations of Givens rotations. Apply one complete iteration of given rotations to this matrix.

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Good Luck!!!

Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2021

Course Title: Algorithms-I
Time: 3 Hours

Course No: **CSE 209**
Full Marks: **60**

Read the question carefully. There are five sections. Each section carries equal marks. Figures in the right margin indicate marks.

1)

SECTION-1

Section No. 1 is based on ***CO1***. There are seven questions.
You have to answer **any six questions** from this section.

a) (i) Define Algorithm. 2
(ii) Differentiate between Big O, Big Omega and Big Theta notation in single sentences.

b) State the worst case and best-case example for the following sorting algorithms. If there is no such scenario, also specifically mention it. 2
(i) Insertion Sort
(ii) Quick Sort

c) Mention the properties of dynamic programming. 2

d) Identify the limitations of binary search. 2
Explain binary search on the give sequence $\{7, 3, 0, -1, -2\}$ for $Key = 1$. If it is impossible to perform binary search, then explain why.

e) Explain divide and conquer paradigm. 2

f) Differentiate between weakly connected component and strongly connected component. 2

g) Name the representation techniques of a graph into computer memory. Also discover the advantages of one technique over the other under different situations. 2

2)

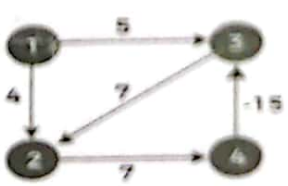
SECTION-2

Section No. 2 is based on ***CO2***. There are four questions.
You have to answer **any three questions out of four** from this section.

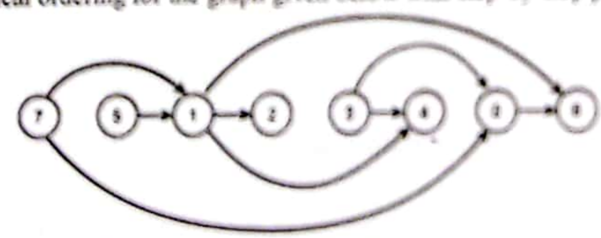
a) Given an array of integers as below: 38, 27, 43, 3, 9, 82, 10, 12, 20, 18 4
Now illustrate the sorting of this array using 2-way and 3-way merge sort and compare their time complexity.
Also find the complexity of these algorithms using big O notation.

b) Explain the Quick sort with the following sequence as example (step by step). 4
$\{4, 6, 2, 8, 1, 5, 3\}$
Also find the complexity of the algorithm using big O notation.

c) Inspect if it is possible to find the shortest path distance from source vertex 1 to destination vertex 4 using Bellmand Ford algorithm (after showing all the iterations). 4



d) Explain Topological ordering for the graph given below with step-by-step procedure. 4



Also find the complexity of your solution using Big O notation.

SECTION-3

Section No. 3 is based on *CO3*. There are four questions.
You have to answer **any three questions out of four** from this section.

a) 4

	1	2	3	4	5	6	7
T	5	4	3	2	7	4	2
D	6	8	9	9	14	15	17

Analyze optimal scheduling strategy to minimize maximum lateness. Also find the complexity of the solution using big O notation.

b) Demonstrate step by step procedure of Heap Sort algorithm (using max heap) for the given sequence $\{5, 6, 3, 1, 4, 2\}$. 4
Also find the complexity of the algorithm using big O notation.

c) Given Text, $X =$ "$BACABACB$" and Text, $Y =$ "$ACBAABAC$". 4
Calculate the LCS between two text X and Y using tabular format. Also formulate a recursive solution and mention the complexity of the solution using big O notation.

d) Given the following two arrays as: 4

Num			rem		
3	4	5	2	3	1

You need to find minimum positive number *X* such that the following is true:

```
x % num[0]     =  rem[0],
x % num[1]     =  rem[1],
.....................
x % num[k-1]   =  rem[k-1]
```

Simulate Chinese remainder theorem (Inverse modulo based implementation) to figure out X.

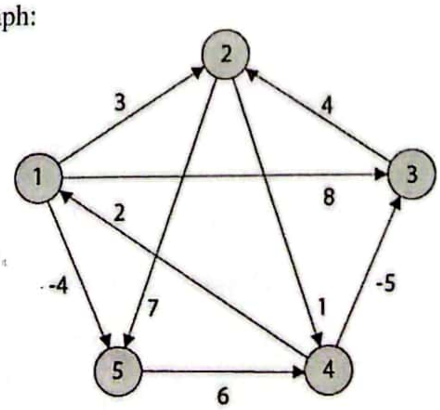
SECTION-4

Section No. 4 is based on *CO4*. There are four questions.
You have to answer **any three questions out of four** from this section.

a) Analyze the application for Huffman coding in the field of computer science within two sentences. Hence, point out the length of the codeword using Huffman coding. 1+3

	a	b	c	d	E	f
Frequency	34	14	12	16	14	10

b) Consider the following graph: 4



(i) Debate "the shortest path distance between every pair of vertices cannot be calculated using Dijktra algorithm".

(ii) Point out an algorithm to calculate the shortest path between all pair of vertices and list out the shortest path distance between all the pair of vertices for the above graph.

c) Explain 0/1 Knapsack problem and find the optimum solution for the given dataset. 4
Capacity of knapsack, K_C=6

Item	Value	Weight
1	12	2
2	10	4
3	9	3
4	6	1

Also find the complexity of the solution using big O notation.

d) A list of different activities with starting and ending times. Calculate the maximum size of non-overlapping activities that can be performed. 4

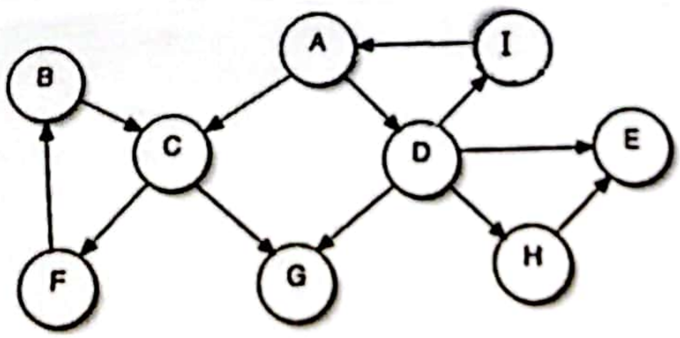
Activities	1	2	3	4	5	6	7	8
Start Time	1	0	1	4	2	5	3	4
Finish Time	3	4	2	6	9	8	5	5

Also find the complexity of the solution using big O notation.

SECTION-5

Section No. 5 is based on *CO5*. There are three questions.
You have to answer **any two questions out of three** from this section.

5) a) Identify different types of edges (forward edge, back edge, tree edge, cross edge, etc) using DFS graph search algorithm. Consider A as starting vertex (node). Also find SCC for the given graph. 3+3



b) Given a set of non-negative integers as set [] = {3,34,4,12,5,2} and a value sum = 9. 6
Find out if there is any subset of the given set with sum equal to the given sum using dynamic programming approach (show the recursion tree also).

c) Given a recurrence relation, $f_n = (2f_{n-1} + 3\, f_{n-2}) \bmod 101$. 3+3
(i) Formulate a solution for any value of n and describe it briefly.
(ii) Calculate the result for n = 40 using matrix exponentiation.

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Good Luck!!!!

Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B. Sc. (Honors) Final Examination, 2021

Course Title: Computer Ethics and Cyber Law
Time: 3 Hours

Course No: **CSE-203**
Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1.

Section-I (CO1)

(Answer any six of the following questions.)

a) Define computer ethics and cyber space. 2

b) Define spyware. Explain how it is being used. 2

c) Explain the approach that we should follow during ethical decision making. 2

d) Point out the term intellectual property encompass. Explain why the companies are so concerned about protecting it. 2

e) Define open-source code. List out the fundamental premise behind its use? 2

f) Define cyberspoofing. State at least one example.

g) List out the punishment in Bangladesh for stealing computer documents, assets or any software's source code from any organization, individual, or from any other means. 2

2.

Section-II (CO2)

(Answer Any Three out of Four)

a) According to Cyber law, how does DDoS attack make huge loss to a computer service? 4

b) Explain why are corporations interested in fostering good business ethics? 4

c) Explain How do codes of ethics, professional organizations, certification, and licensing affect the ethical behavior of IT professionals? 4

d) "A code of polite, thoughtful, or respectful behaviors to be followed while using networks, including the network" is known as Nettiquette. Can you develop a small sample Nettiquette for school going students? 4

3.

Section-III (CO3)

(Answer Any Three out of Four)

a) List out different types of actions that the Controller of Certifying Authorities (CCA) may perform in case of cyber security. Exemplify with a scenario where CCA may take a role to perform different functions according to the Law of Bangladesh. 4

b) Point the four most common types of software product liability claims and explain them briefly. Identify the actions must plaintiffs and defendants take to be successful. 4

c) A musician created a new song. She will be protected from others using her song because of ____________________. 4

(i) Copyright Law
(ii) Spoofing;
(iii) Netiquette
(iv) Fair use.

Explain in detail of your choice in context with Bangladesh Law based on the above-mentioned points.

d) Explain whistle blowing. Point out the ethical issues associated with it. Illustrate the steps of effective whistle blowing. 4

4.

Section-IV (CO4)

(Answer Any Three out of Four)

a) Justify the code of ethics for information technology professionals. 4

b) Estimate some characteristics of common computer criminals, including their objectives, available resources, willingness to accept risk, and frequency of attack? 4

c) Evaluate the most significant legal issue in computer forensics amongst the following 4
   (i) Preserving Evidence;
   (ii) Admissibility of Evidence;
   (iii) Discovery of Evidence or
   (iv) Seizing Evidence.
   Justify your answer in detail in contrast with other options.

d) Estimate the ethical issues software manufacturers may face during trade-offs between project schedules, project costs, and software quality. 4

5.

Section-V (CO5)

(Answer Any Two out of Three)

a) Based on incident prioritization, which one of the following incidents should have first priority (Priority 1) and so on? 6
   (i) ***GatorLink*** account compromised and being used to send spam.
   (ii) Multifunction printer/fax/scanner servicing a department stops functioning.
   (iii) ***JUFL*** is down; hacking/compromise of critical University Financial system leading to service unavailability/disclosure of restricted data.
   (iv) ***ELearning*** is down but during spring break; AP Pay cycle will not run during the beginning of a pay period.
   (v) Videoconferencing via ***Polycom*** is unavailable for a specific conference.
   What would be your choice as a Computer Emergency Response Team (CERT) member while you face all these issues simultaneously in your organization? Critisize your answer of priority.

b) Do you think that law enforcement agencies should be able to use advanced surveillance technology, such as surveillance cameras combined with facial recognition software? Justify your answer briefly. 6

c) Evaluate the steps that an IT professionals do to ensure that the projects they lead meet the client's expectations and do not lead to charges of fraud, fraudulent misrepresentation, and breach of contract. 6

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Good Luck!!!

Jahangirnagar University
Department of Computer Science and Engineering
2^{nd} Year 1^{st} Semester B.Sc. (Hons.) Final Examination -2021

Course Title: Mathematics III (Vector, Complex Variable, Fourier Analysis and Laplace Transformation) **Time:** 3 Hours
Course No: MATH 201 **Full Marks:** 60

Each section carries equal marks. Figures in the right margin indicate marks.
Answer any Six (06) questions out of Eight (08) questions.

1. a. Define the gradient of a Scalar, the Divergence of a vector, and the curl of a vector. If $\phi(x,y,z) = 3x^2y - y^3z^2$ find $\underline{\nabla}\phi$ at the point $(1,-2,-1)$. 5

   b. Show that $\underline{\nabla} r^n = nr^{n-2}\underline{r}$, where $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$. 5

2. a. State Green's theorem, Stokes theorem and the Gauss divergence theorem. 5

   b. Find the total work done in moving a particle in a force field given by 5
   $\underline{F} = 3xy\underline{i} - y^2\underline{j}$ along the curve $y = 2x^2$. $-12x^2y$

3. a. If $f(z)$ is analytic at every point on and within the simply closed curve C where a is within C then show that 4

$$\int_C \frac{f(z)}{z-a}\,dz = 2\pi i\, f(a)$$

Course Title: Mathematics III (Vector, Complex Variable, Fourier Analysis and Laplace Transformation) **Time:** 3 Hours
Course No: MATH 201 **Full Marks:** 60

Each section carries equal marks. Figures in the right margin indicate marks.
Answer any Six (06) questions out of Eight (08) questions.

1. a. Define the gradient of a Scalar, the Divergence of a vector, and the curl of a vector. If $\phi(x,y,z) = 3x^2y - y^3z^2$ find $\underline{\nabla}\phi$ at the point $(1,-2,-1)$. 5

   b. Show that $\underline{\nabla} r^n = nr^{n-2}\underline{r}$, where $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$. 5

2. a. State Green's theorem, Stokes theorem and the Gauss divergence theorem. 5

   b. Find the total work done in moving a particle in a force field given by $\underline{F} = 3xy\underline{i} - y^2\underline{j}$ along the curve $y = 2x^2$. 5 $-12x^2y$

3. a. If $f(z)$ is analytic at every point on and within the simply closed curve C where a is within C then show that 4

$$\int_C \frac{f(z)}{z-a}dz = 2\pi i\, f(a)$$

   b. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z^2 - 9z + 20}dz$; where $C: z = 6e^{i\theta}; 0 \le \theta \le 2\pi$. 3

   c. Evaluate $\oint_C \frac{e^{2z}}{(z-1)^4}dz$; where C is the circle $|z| = 3$. $2\pi i e^2$ 3

4. a. State and proof Taylors Theorem. 5

   b. Expand $f(z) = \dfrac{1}{(z+1)(z+3)}$ in a Laurent series valid for: 5

      i. $|z| > 3$
      ii. $0 < |z+1| < 2$
      iii. $|z| < 1$

5. a. Find the residues of $f(z) = e^z \operatorname{cosec}^2 z$ at all its poles in the finite plane. 3

   b) Evaluate $\int_0^\infty \frac{dx}{x^4 + a^4}$; $a > 0$. 5

   c) State Laurent Series. 2

a. Find the Fourier series expansion of $f(x)=\begin{cases} x-\pi, & -\pi<x<0 \\ \pi-x, & 0<x<\pi \end{cases}$. 5

b. Find the Fourier Transform of $f(x)=\begin{cases} 1-x^2, & |x|<1 \\ 0, & |x|>1 \end{cases}$. 5

7. a. How do you define an analytic function? Obtain Cauchy-Riemann equations. Express, giving all detail, these equations in polar form. 5

b. Determine whether $u(x,y)=e^{-2xy}\, Sin(x^2-y^2)$ is harmonic or not. If so, find the conjugate harmonic function of u. 5

a. Define Laplace transform. Find the Laplace transform of the function $F(t)=t^n e^{at}$. 4

b. Solve the following differential equations by using Laplace transform 6

i. $\dfrac{dx}{dt}+5x=e^{2t}, \quad x(0)=1$

ii. $\dfrac{d^2y}{dt^2}+y=3\sin(2t), \quad y(0)=3, y'(0)=1.$

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Good Luck!!!

Department of CSE
Jahangirnagar University
Savar, Dhaka-1342.

Course No: Math 201 | Class Test-02 | Time: 35 Minutes | Marks: 30

1. Using Cauchy's residue theorem evaluate $\int_0^\infty \frac{dx}{x^4+81}$.

2. If $\underline{a} = x^2y\hat{\imath} - 2xz\hat{\jmath} + 2yz\hat{k}$, find $\underline{\nabla} \times (\underline{\nabla} \times \underline{a})$ at the point $(1, -1, 1)$.

3. Show that $\underline{\nabla} r^n = nr^{n-2}\underline{r}$, where $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$.

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Course No: Math 201 **Class Test-01** **Time: 45 Minutes** Marks: 30

1. (a) Prove that the function $u = e^{-x}(x\sin y - y\cos y)$ is harmonic.
   (b) Find a function v such that $f(z) = u + iv$ is analytic.
   (c) Express $f(z)$ in terms of z.

2. Evaluate $\oint\limits_C \frac{e^{2z}}{(z-2)^4}dz$; where C is the circle (a) $|z| = 3$, (b) $|z| = \frac{1}{2}$.

3. Expand $f(z) = \frac{z}{(z-1)(2-z)}$ in a Laurent series valid for
   (a) $|z-1| > 1$
   (b) $0 < |z-2| < 1$.

Department of Computer Science and Engineering
Jahangirnagar University, Savar, Dhaka
Tutorial no.1

Course Code: CSE 203 — Title: Computer Ethics and Cyber Law
Time: 30 mins. — Full Marks:15

1. What is computer ethics? Describe Ten Commandments of computer ethics? 5
2. What is Cyber Space? Describe the Code of ethics for information technology professionals. 5
3. Describe why are corporations interested in fostering good business ethics? 5
4. What relationships must an IT professional manage, and what key ethical issues can arise in each? 5

Department of Computer Science and Engineering
Jahangirnagar University, Savar, Dhaka
Tutorial no.2

Course Code: CSE 203 Title: Computer Ethics and Cyber Law
Time: 30 mins. Full Marks:10
Answer Any Two of the following Questions

1. Describe what are some characteristics of common computer criminals, including their objectives, available resources, willingness to accept risk, and frequency of attack? 5
2. (a) Describe what actions must be taken in response to a security incident? 3
   (b) When does zero attack take place? 2
3. What are the two fundamental forms of data encryption, and how does each work? 5
4. What relationships must an IT professional manage, and what key ethical issues can arise in each? 5

Department of Computer Science and Engineering
Jahangirnagar University, Savar, Dhaka

Tutorial no.3

Course Code: CSE 203
Time: 30 mins.

Title: Computer Ethics and Cyber Law
Full Marks:15

1. What is identity theft, and what techniques do identity thieves use? What are the various strategies for consumer profiling and the associated ethical issues? 5

2. What are the strengths and limitations of using copyrights, patents, and trade secret laws to protect intellectual property? 5

3. What are the two fundamental forms of data encryption, and how does each work? 5

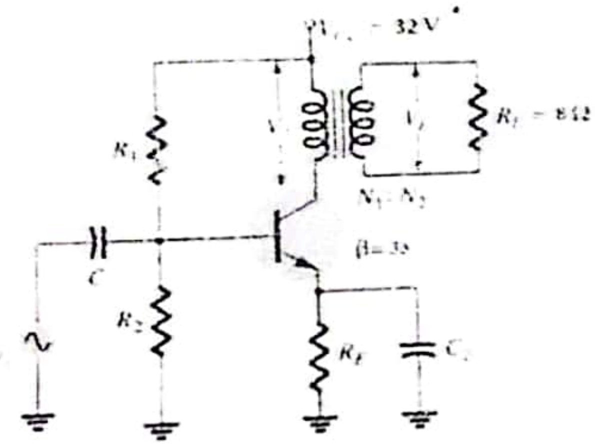
1. **Draw and describe the configuration of Quasi-Complementary Push-Pull Amplifier, Transformer-Coupled Push-Pull Class B Amplifier.**
2. **Describe the operation and application of LCD.**
3.

Consider the following transformer coupled class-A Power Amplifier circuit shown in figure Q2, the maximum output power of the amplifier is equal to 8 Watts and its I_{BQ} =15.2 mA.

i- Determine the required transformer turn ratio N1:N2.

ii- Calculate it maximum input D.C. power and find the amplifier efficiency.

iii- What are the values of V_{LP} and I_{LP}?



Jahangirnagar University
Department of Computer Science and Engineering

2nd year 1st semester 2021

CT-1

Course nc: CSE-207

Time: 40min | Marks:10

1.	Demonstrate the characteristics of bandpass filter.	2
2.	Draw and describe how basic instrumentation amplifier works?	2
3.	What is slew rate?	2
4.	Mention the works of Zero Drift Amplifier.	2
5.	Calculate the output voltage for this circuit.	2